EECS 16A Designing Information Devices and Systems I
Homework 9

Submission Format
Your homework submission should consist of **one** file.

- `hw9.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. **Reading Assignment**

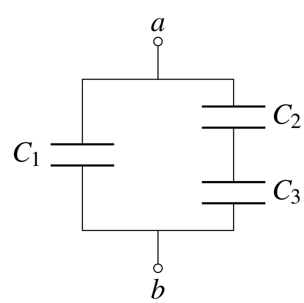
   For this homework, please read Notes 16 and 17. Note 16 will provide an introduction to capacitors (a circuit element which stores charge), capacitive equivalence, and the underlying physics behind them. Sections 17.1 - 17.2 in Note 17 will provide an overview of the capacitive touchscreen and how to measure capacitance. 17.3 will introduce you to the op-amp and how it can be used as a comparator. 17.4 will discuss implementing a capacitive touchscreen with an op-amp comparator.

   (a) Describe the key ideas behind how a capacitor works. How are capacitor equivalences calculated? Contrast this with how we calculate resistor equivalences.

   (b) Consider the capacitive touchscreen. Describe how it works, and compare and contrast it to the resistive touchscreens we have seen in previous lectures and homeworks.

2. **Equivalent Capacitance**

   (a) Find the equivalent capacitance between terminals a and b of the following circuit in terms of the given capacitors C_1, C_2, and C_3. Leave your answer in terms of the addition, subtraction, multiplication, and division operators **only**.



   (b) Find and draw a capacitive circuit using three capacitors, C_1, C_2, and C_3, that has equivalent capacitance of

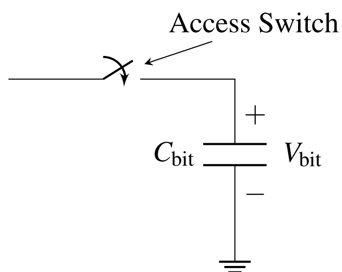
$$\frac{C_1(C_2+C_3)}{C_1+C_2+C_3}$$

3. Dynamic Random Access Memory (DRAM)

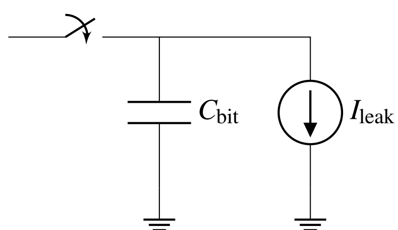
Nearly all devices that include some form of computational capability (phones, tablets, gaming consoles, laptops, ...) use a type of memory known as Dynamic Random Access Memory (DRAM). DRAM is where the "working set" of instructions and data for a processor is typically stored, and the ability to pack an ever increasing number of bits on to a DRAM chip at low cost has been critical to the continued growth in computational capability of our systems. For example, a single DRAM chip today can store $>$ 8 billion bits and is sold for $\approx$ \$3-\$5.

At the most basic level and as shown below, every bit of information that DRAM can store is associated with a capacitor. The amount of charge stored on that capacitor (and correspondingly, the voltage across the capacitor) determines whether a "1" or a "0" is stored in that location.

Single DRAM Bit Cell



In any real capacitor, there is always a path for charge to "leak" off the capacitor and cause it to eventually discharge. In DRAMs, the dominant path for this leakage to happen is through the access switch, which we will **model as a leakage to ground**. The figure below shows **a model of this leakage**:



Fun Fact: This leakage is actually responsible for the "D" in "DRAM" – the memory is "dynamic" because after a cell is "written" by storing some charge onto its capacitor, if you leave the cell alone for too long, the value you wrote in will disappear because the charge on the capacitor leaked away.

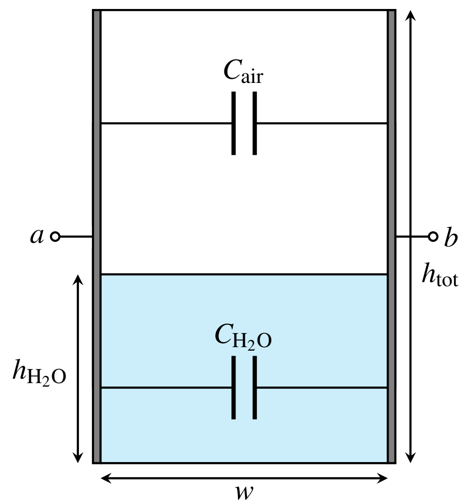
Let's now try to use some representative numbers to compute how long a DRAM cell can hold its value before the information leaks away. Let $C_{\text{bit}} = 28\,\text{fF}$ (note that $1\,\text{fF} = 1 \times 10^{-15}\,\text{F}$) and the capacitor be initially charged to $1.2\,\text{V}$ to store a "1." V_{bit} must be $> 0.9\,\text{V}$ in order for the circuits outside of the column to properly read the bit stored in the cell as a "1."

What is the maximum value of I_{leak} that would allow the DRAM cell retain its value for $> 1\,\text{ms}$?

Hint: Start by writing out the equations you know about charge and current related to the capacitor. Note here that the current source is discharging the capacitor.

4. It's finally raining!

A lettuce farmer in Salinas Valley has grown tired of weather.com's imprecise rain measurements. Therefore, they decided to take matters into their own hands by building a rain sensor. They placed a rectangular tank outside and attached two metal plates to two opposite sides in an effort to make a capacitor whose capacitance varies with the amount of water inside.

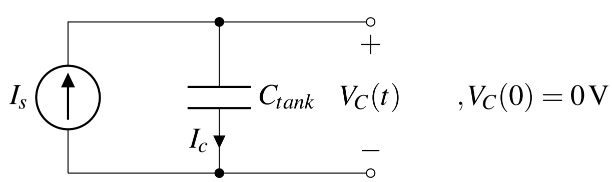


The width and length of the tank are both w (i.e., the base is square) and the height of the tank is h_{tot}.

(a) What is the capacitance between terminals a and b when the tank is full? What about when it is empty? *Note:* the permittivity of air is ε, and the permittivity of rainwater is 81ε.

(b) Suppose the height of the water in the tank is $h_{\text{H}_2\text{O}}$. Modeling the tank as a pair of capacitors in parallel, find the total capacitance between the two plates. Call this capacitance C_{tank}.

(c) After building this capacitor, the farmer consults the internet to assist them with a capacitance-measuring circuit. A fellow internet user recommends the following:



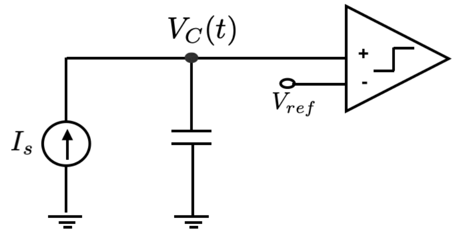
In this circuit, C_{tank} is the total tank capacitance that you calculated earlier. I_s is a known current supplied by a current source.

The suggestion is to measure V_C for a brief interval of time, and then use the difference to determine C_{tank}.

Determine $V_C(t)$, where t is the number of seconds elapsed since the start of the measurement. You should assume that before any measurements are taken, the voltage across C_{tank}, i.e. V_C, is initialized to $0\,\text{V}$, i.e. $V_C(0) = 0$.

(d) Using the equation you derived for $V_C(t)$, describe how you can use this circuit to determine C_{tank} and $h_{\text{H}_2\text{O}}$.

(e) However, after spending some time thinking about different ways of measuring this capacitance you came up with a better idea. You decided to use the circuit proposed in part (c) along with a comparator, as show in the figure below. What you are basically interested in, is the time T_1 needed for V_c to reach V_{ref}. In order to measure time you use a timer. When voltage V_c becomes larger than V_{ref}, the comparator flips its value and you stop the timer. How would you measure in that case the value of the capacitance?



5. Capacitive Touchscreen

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The model for a capacitive touchscreen can be seen in Figure 1. See Table 1 for values of the dimensions. The green area represents the contact area of the finger with the top insulator. It has dimensions w_2 x d_1, where w_2 is the horizontal width of the finger contact area and d_1 is the depth (into the page) of the finger contact area. The top metal (red area) has dimensions $w_1 \times d_1$. The bottom metal (grey area) has dimensions $w \times d_2$, where w is larger than both w_1 and w_2.

Table 1: Touchscreen Dimension Values

d_1	10 mm
d_2	1 mm
t_1	2 mm
t_2	4 mm
w_1	1 mm
w_2	2 mm

(a) Draw the equivalent circuit of the touchscreen that contains the nodes F, E_1, and E_2 when: (i) there no finger present; and (ii) when there is a finger present. Express the capacitance values in terms of C_0, C_{F-E1}, and C_{F-E2}.

Hint: Note that node F represents the finger. When there is no touch node F would be non-existent. Hint: Treat E_1 as the "top node", E_2 as the "bottom node", and the finger F as an intermediate node when present.

(b) What are the values of C_0, C_{E1-F}, and C_{F-E2}? Assume that the insulating material has a permittivity of $\varepsilon = 4.43 \times 10^{-11} F/m$ and that the thickness of the metal layers is small compared to t_1 (so you can ignore the thickness of the metal layers).

(c) What is the difference in effective capacitance between the two metal plates (nodes E_1 and E_2) when a finger is present?

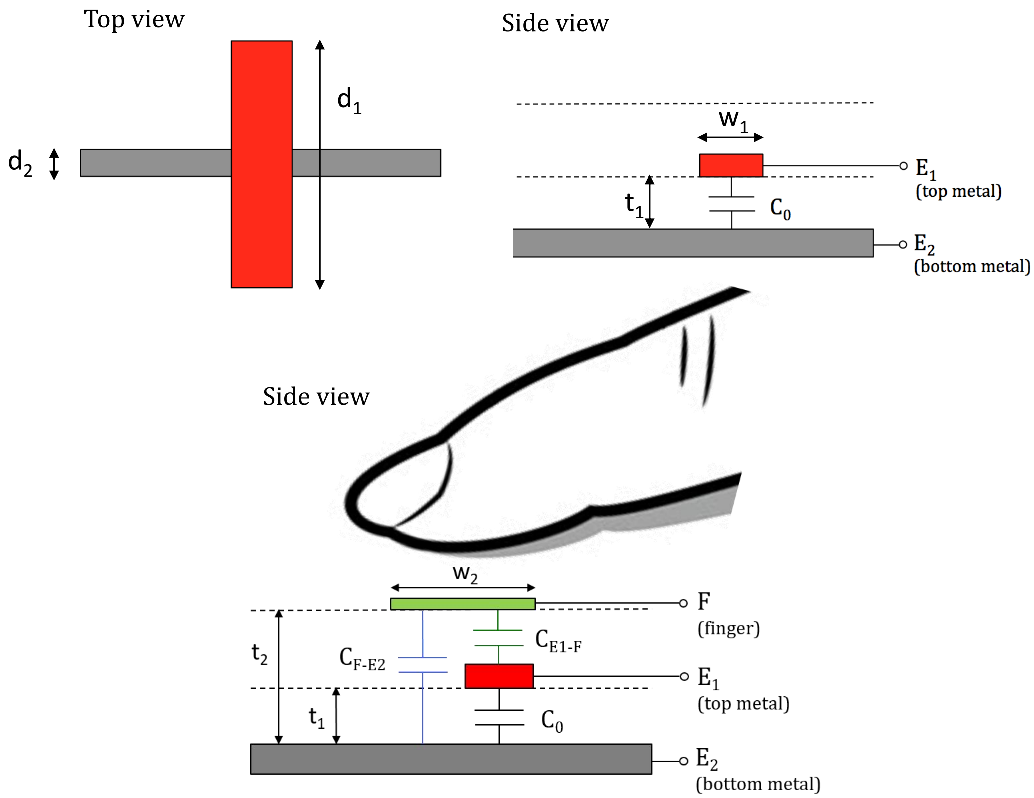

Figure 1: Model of capacitive touchscreen.

6. Super-Capacitors

In order to enable small devices for the "Internet of Things" (IoT), many companies and researchers are currently exploring alternative means of storing and delivering electrical power to the electronics within these devices. One example of these are "super-capacitors" - the devices generally behave just like a "normal" capacitor but have been engineered to have extremely high values of capacitance relative to other devices that fit in to the same physical volume. They can function as a power supply for low power applications such as IoT devices and have the advantage that they can be charged and discharged many times without losing maximum charge capacity. That property makes super-capacitors suitable to store energy from intermittent power sources such as those from energy harvesting. Suppose you are tasked with designing a power supply with a super-capacitor in an IoT device.

(a) Assuming that your electronic device (load) can be modeled as a constant current source with a value of i_{load}, draw circuit models for your device using super-capacitors as the power supply with the following configurations:

- Config 1: a single super-capacitor as the power supply
- Config 2: two super-capacitors stacked in series as the power supply
- Config 3: two super-capacitors connected in parallel as the power supply

(b) If each super-capacitor is charged to an initial voltage v_{init} and has a capacitance of C_{sc}, for each of the three configurations above, write an expression for the voltage supplied to your electronic device as a function of time after the device has been activated (i.e. connected to the super-capacitor(s)).

(c) Now let's assume that your electronic device requires some minimum voltage v_{min} in order to function properly. For each of the three super-capacitor configurations, write an expression for the lifetime of the device.

(d) Assume that a single super-capacitor doesn't provide you sufficient lifetime and so you have to spend the extra money (and device volume) for another super-capacitor. You consider the two following configurations:

- Config 2: two super-capacitors stacked in series
- Config 3: two super-capacitors connected in parallel

When is Config 3 (parallel) better than Config 2 (series)? Your answer should involve conditions on v_{init} and v_{min}.

(e) Calculate the amount of energy delivered by the super-capacitors in Config 3 (parallel) over the device's lifetime.

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.